WELCOME

Network Time Protocol

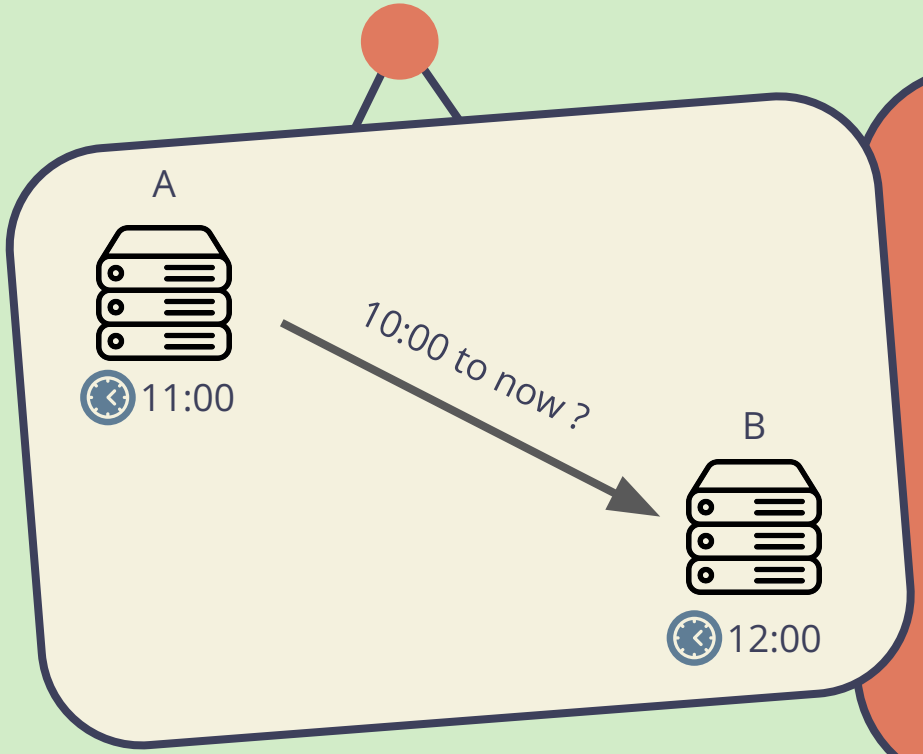
NTP

Brief History

- NTP was one of the first internet protocols.
- NTPv0 was first implemented in 1985.
- In 1992, stripped-down version of NTP, Simple NTP (SNTP) was introduced.
- In 2010 Version 4 was introduced, It has security improvements and support for IPv6.
- NTPv4 is the current version.

Why Computers Need Time?

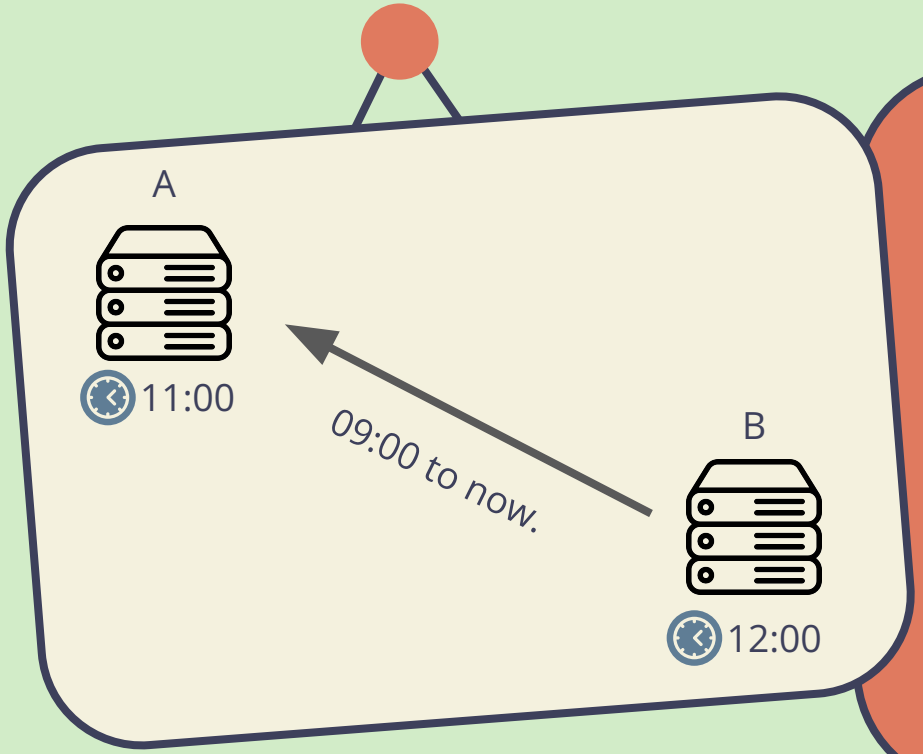
To Synchronize and Coordinate Between Them



What would happen when time between servers are not the same?

- A wants last hour data from B
- B sends last two hours data to A
- A performs calculations on B's data

At this point, the results would be wrong and that's not good.



What would happen when time between servers are not the same?

- A wants last hour data from B
- B sends last two hours data to A
- A performs calculations on B's data

At this point, the results would be wrong and that's not good.

So, What is NTP?

Purpose

It is designed to synchronize computer clocks across networks

Time Source

Highly accurate sources can be obtained from atomic clocks

Public Servers

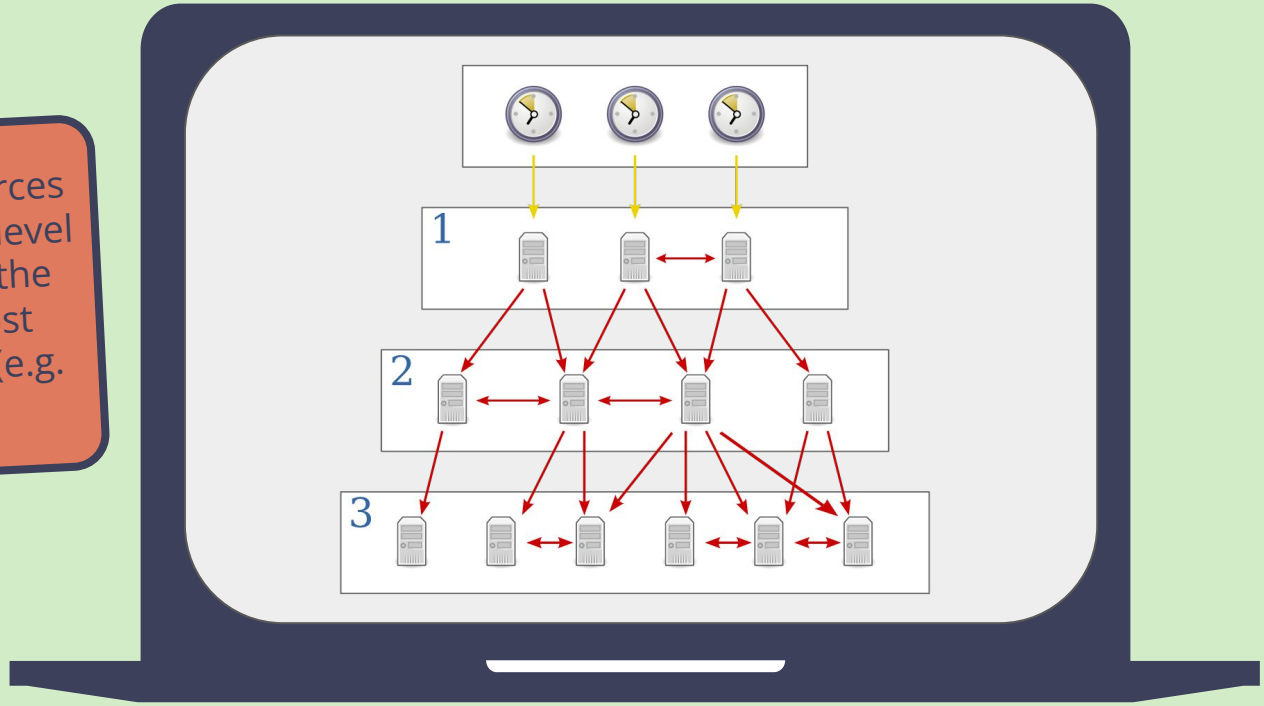
Free public NTP servers can be access via the Internet

Stratum

The hierarchical layer used by NTP for distributing time sources

Clock Strata

NTP organizes time sources into strata. Where each level of stratum represents the distance from the most accurate time sources (e.g. atomic clocks)



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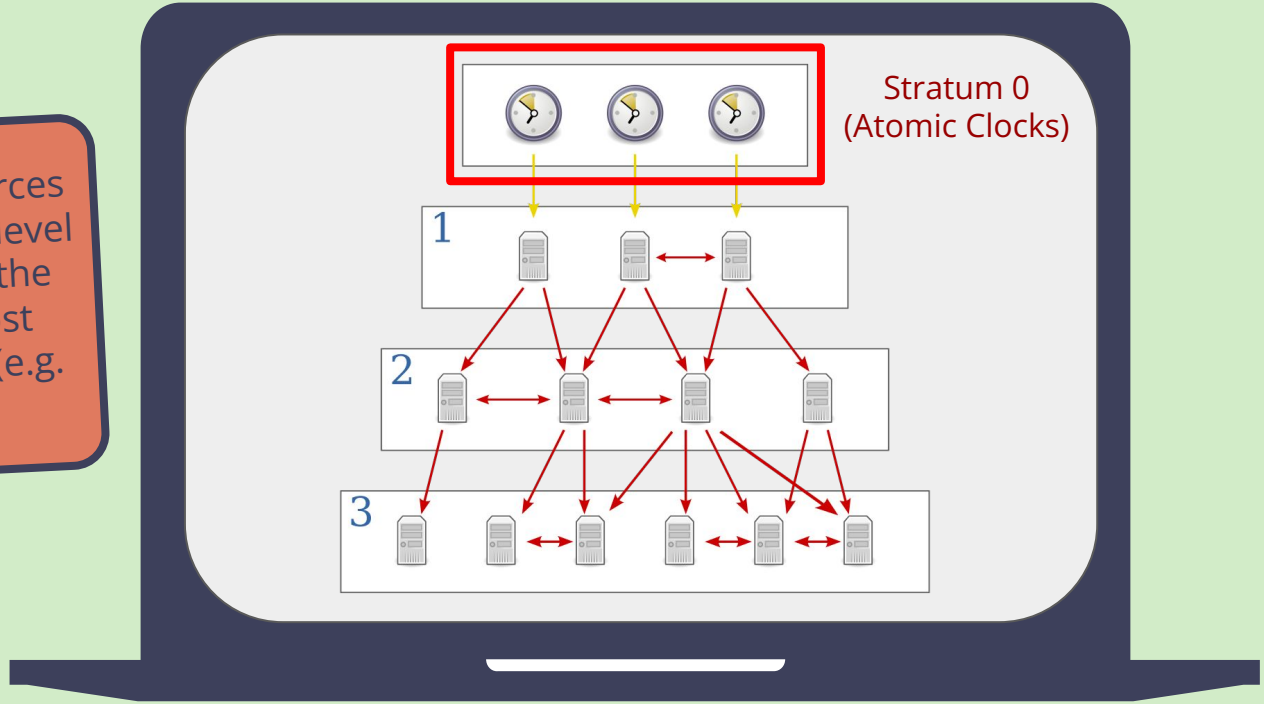

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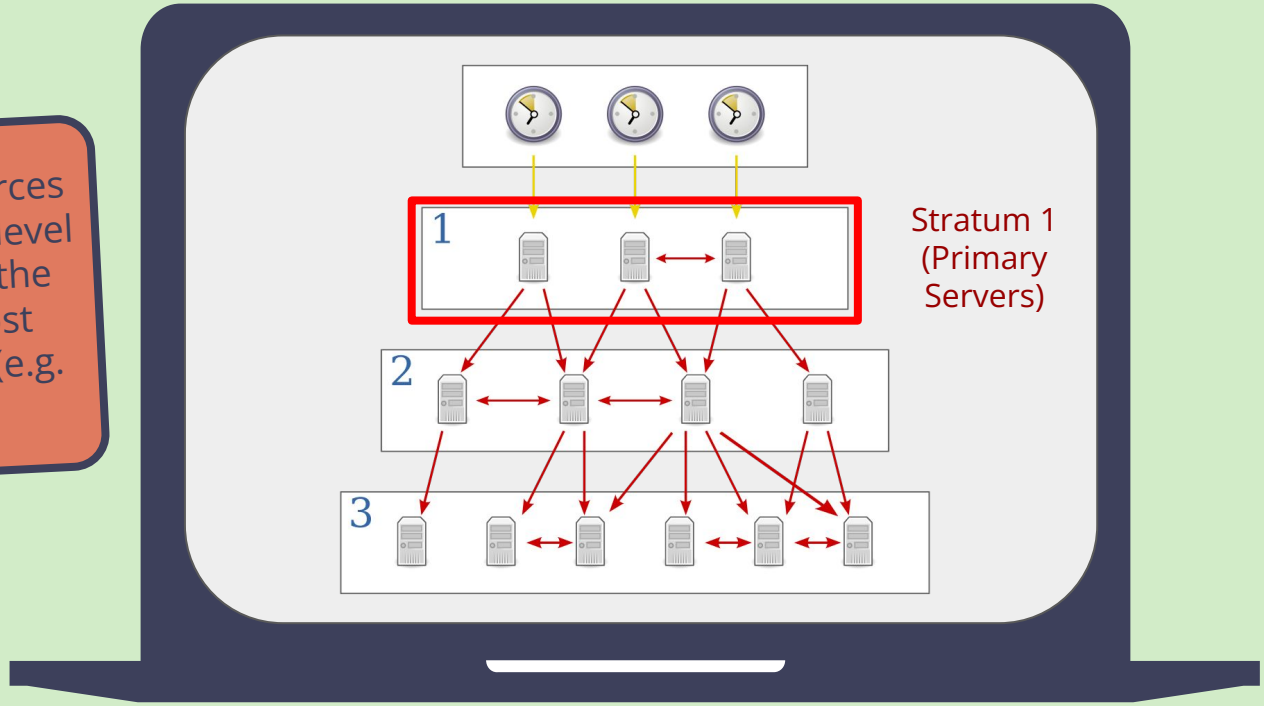

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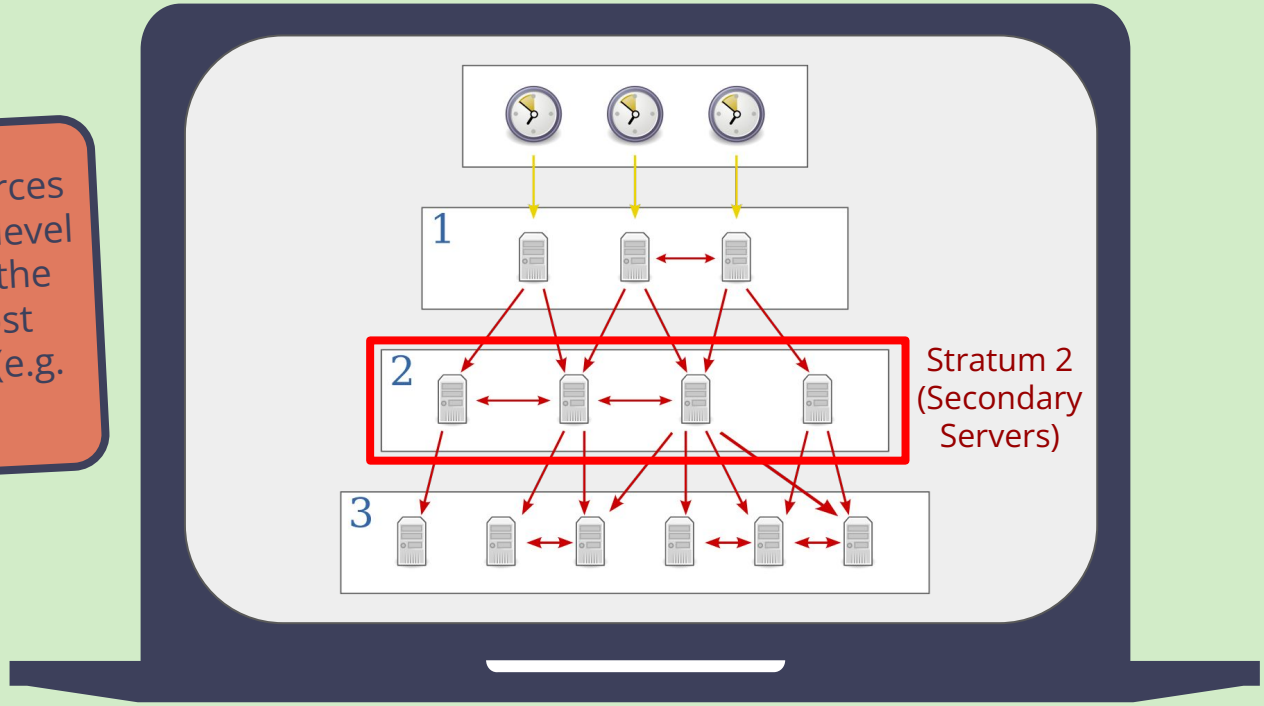

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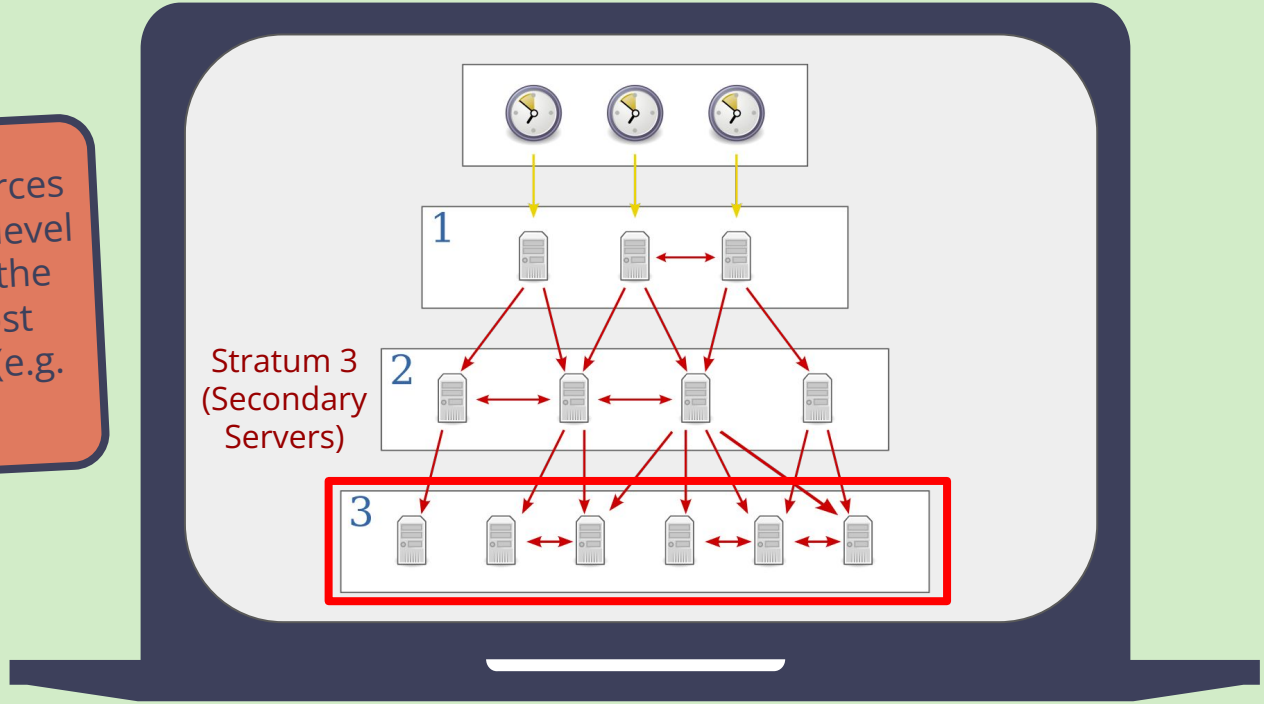
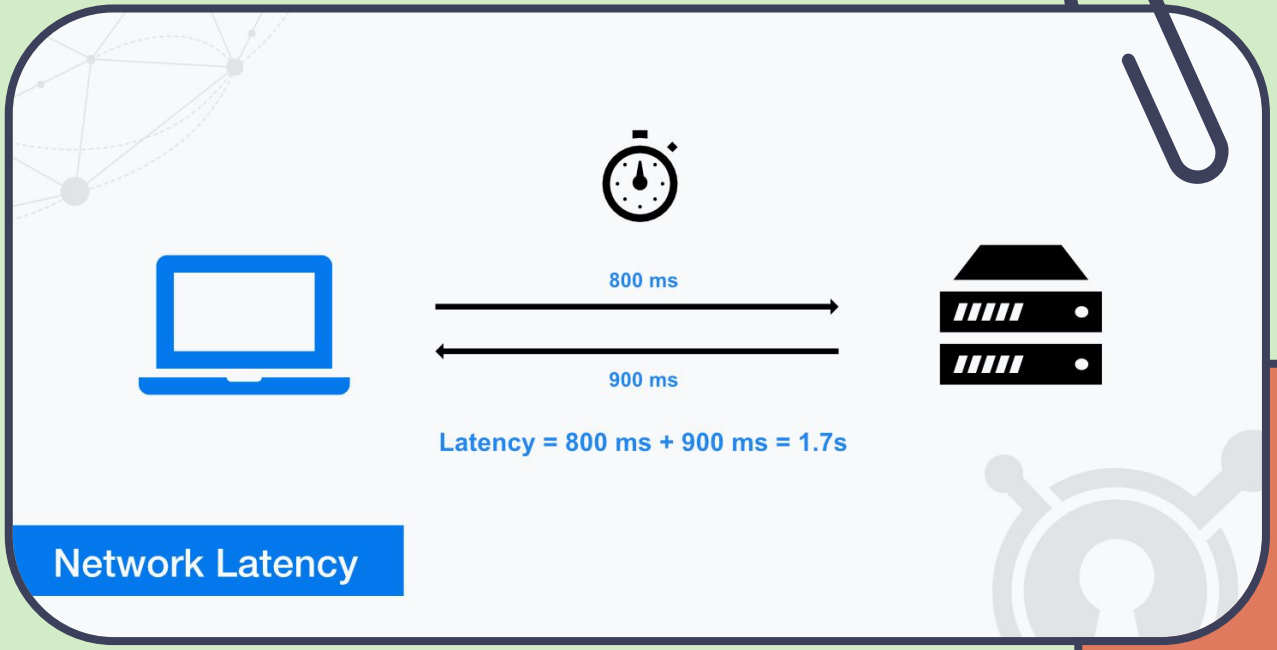




Clock synchronization algorithms help deal with the latency

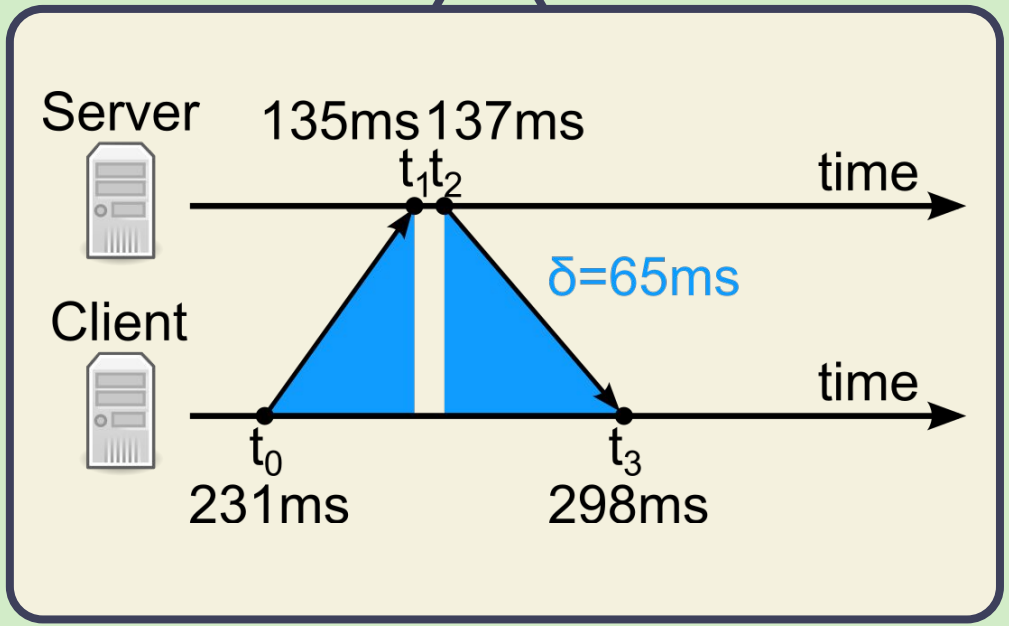
Synchronization algorithm

Offset

$$\theta = \frac{(t_1 - t_0) + (t_2 - t_3)}{2},$$

Round-trip Delay

$$\delta = (t_3 - t_0) - (t_2 - t_1)$$



Time offset θ is positive when client is slower than server and negative when it is faster.

What is SNTP?

Simple Network Time Protocol (SNTP) was defined in 1992 to make NTP more simpler

NTP vs SNTP

Accuracy

SNTP has less accuracy and precision than NTP. which means SNTP is faster to compute

Complexity

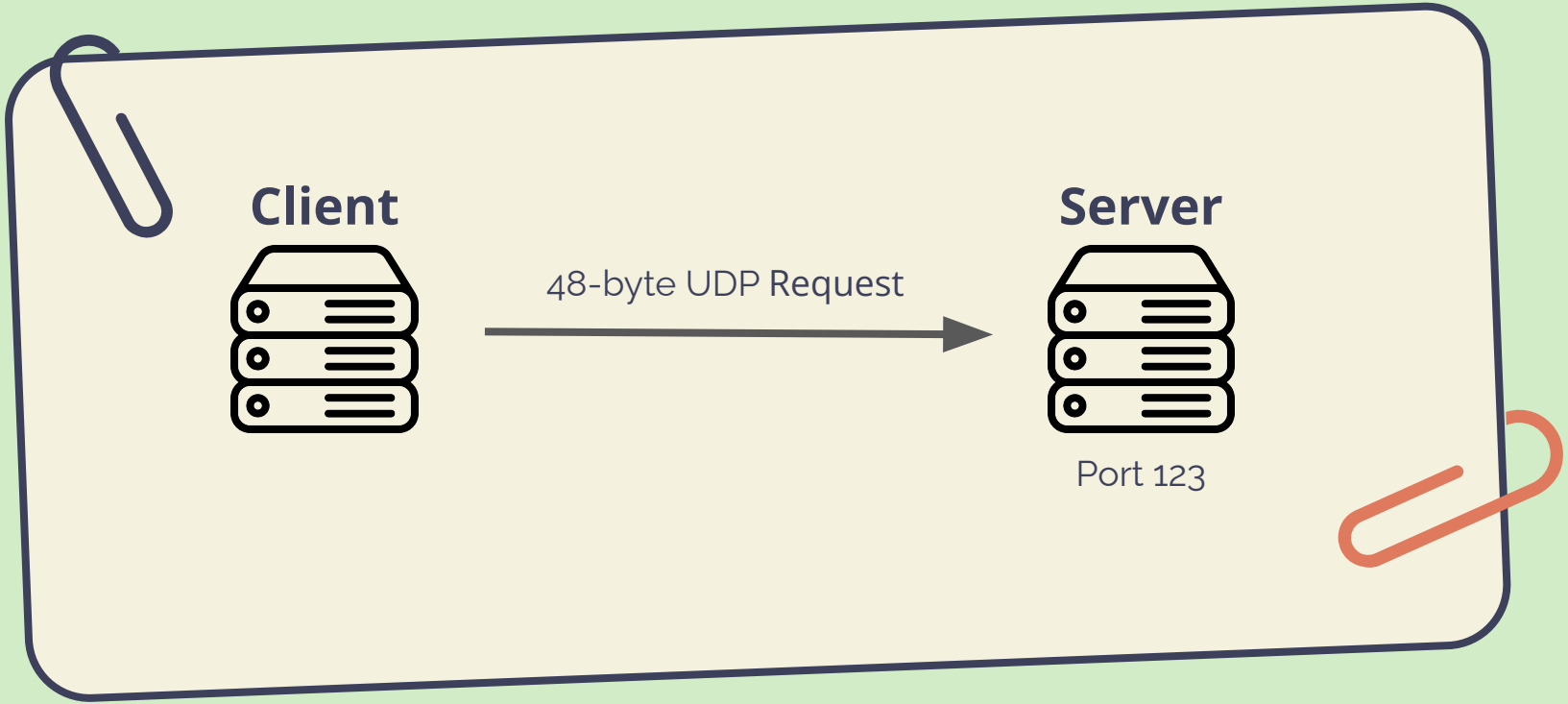
SNTP use simpler algorithm and has less features, hence easier to implement

Resource

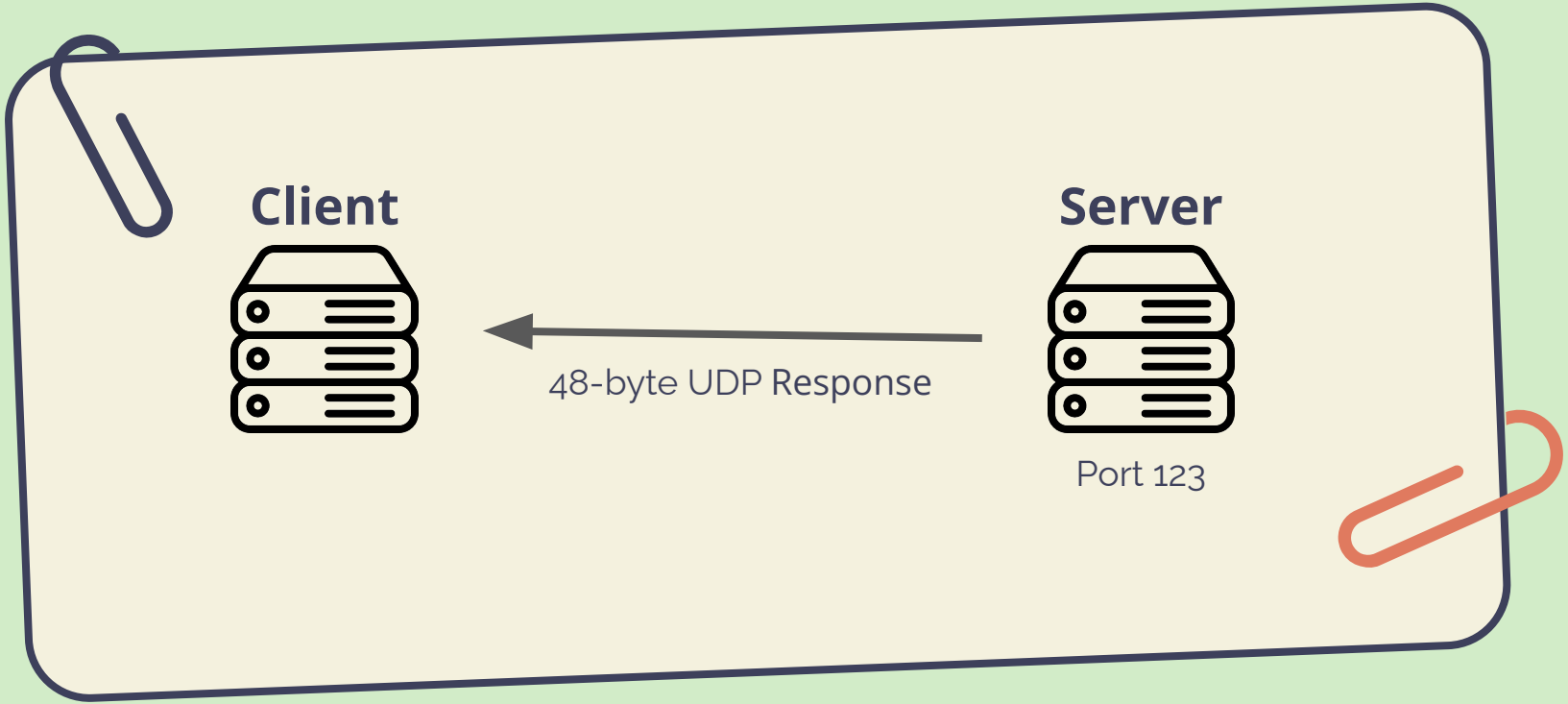
Thanks to its less accuracy and complexity, SNTP requires fewer resources than NTP to run.

Even Though SNTP has less features than NTP, being a subset of NTP, it’s also compatible with NTP

SNTP Request



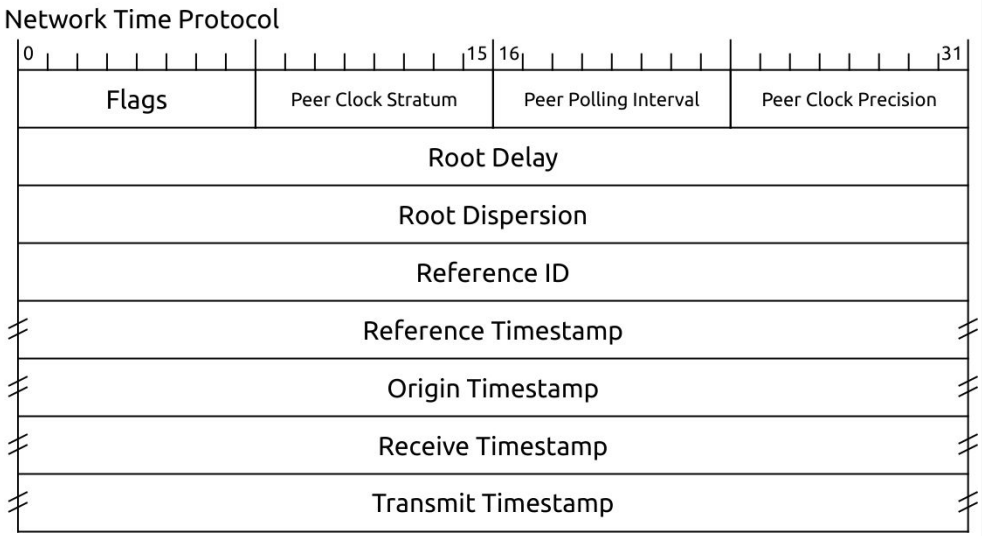
SNTP Response



SNTP Message Format

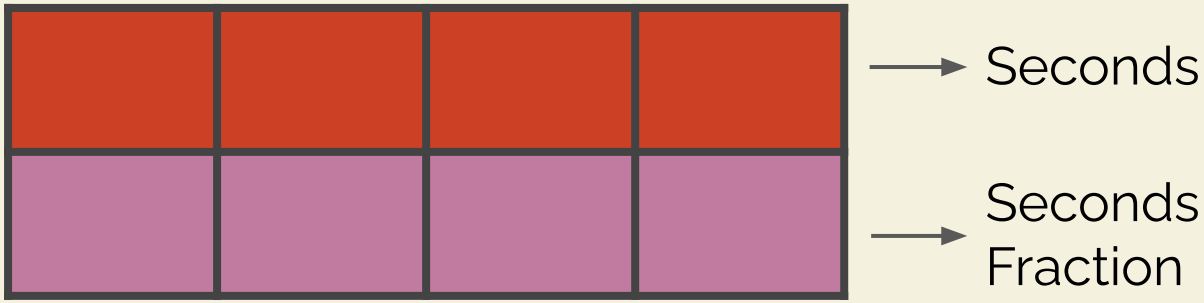
Flags (8):

- Leap Indicator (2):
  - 0 = None
  - 1 = Leap +1
  - 2 = Leap -1
  - 3 = Unsynchronized
- Version Number (3):
  - Current version is 4
- Mode (3):
  - 3 = Client
  - 4 = Server



Clients only need to set Version Number to 4 and Mode to 3. The rest can be 0.

Timestamp Format



Timestamp Format

E8	83	78	AB	= 3900930219
3B	88	06	D9	= 998770393

3900930219 + 998770393/2^32 = 3900930219.232544353

Timestamp Format

E8	83	78	AB	= 3900930219
3B	88	06	D9	= 998770393

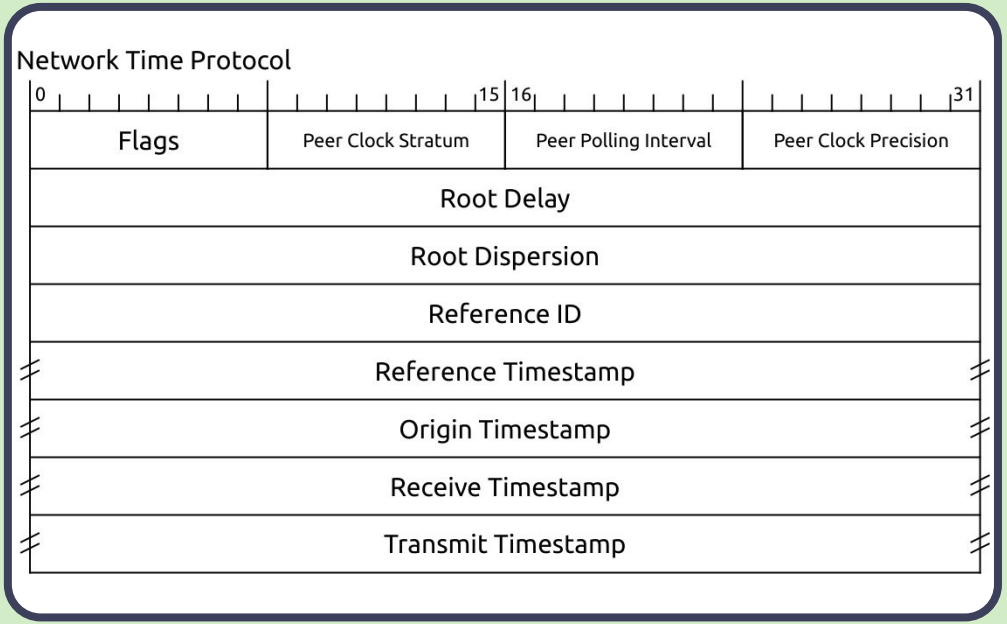
3900930219 + 998770393/2^32 = 3900930219.232544353

Timestamp: Aug 13, 2023 15:43:39.232544353 UTC

Second 0 is Jan 1, 1900 00:00:00.00 UTC

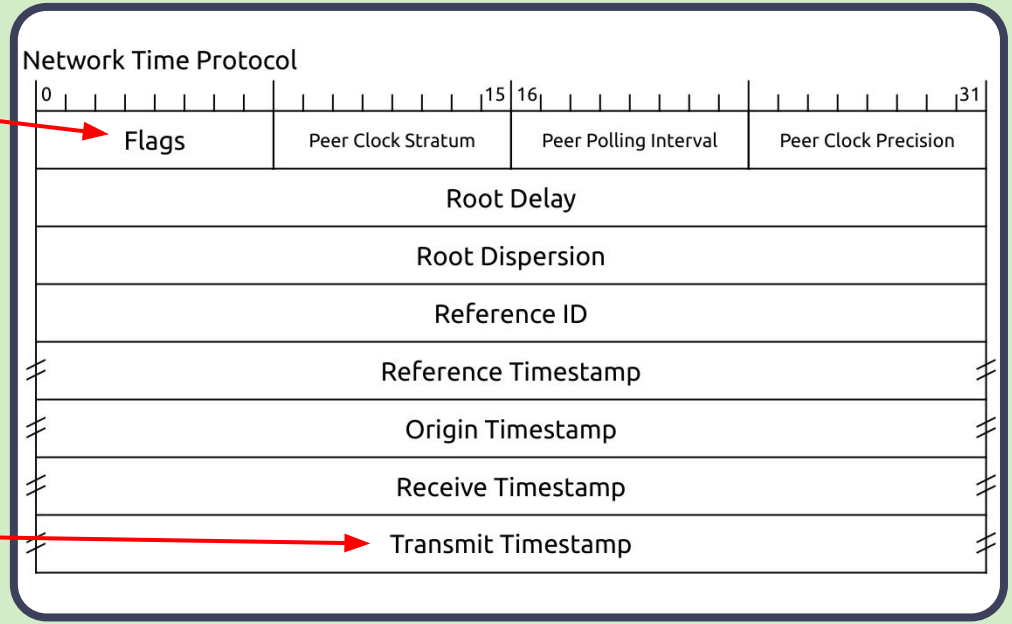
Client to Server Data Frame

```
23 00 00 00
00 00 00 00
00 00 00 00
00 00 00 00
00 00 00 00
00 00 00 00
00 00 00 00
00 00 00 00
00 00 00 00
00 00 00 00
E8 83 78 AB
3B 88 06 D9
```



Client to Server Data Frame

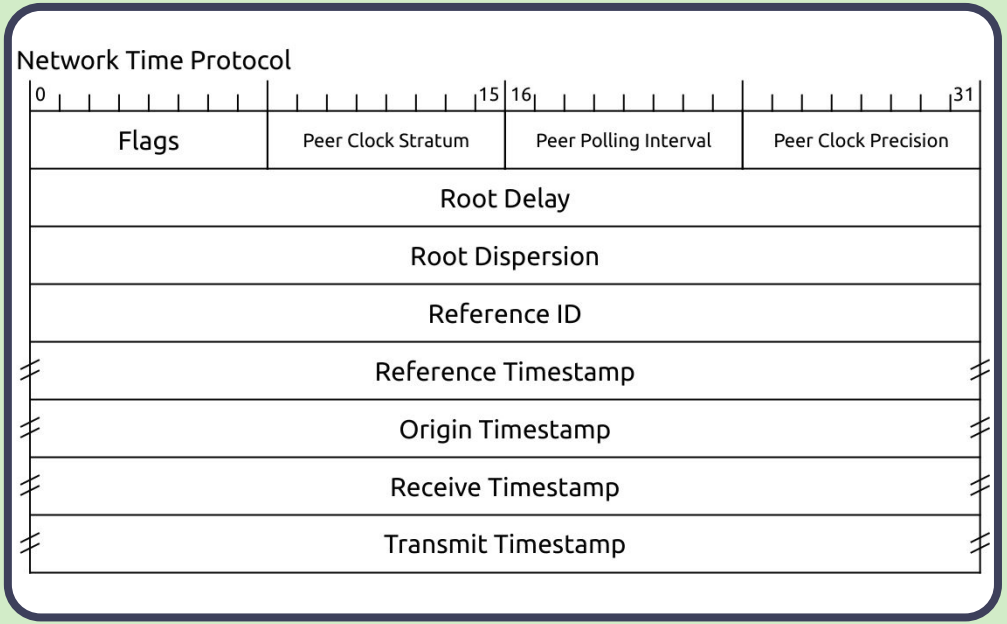
```
23 00 00 00
00 00 00 00
00 00 00 00
00 00 00 00
00 00 00 00
00 00 00 00
00 00 00 00
00 00 00 00
00 00 00 00
00 00 00 00
E8 83 78 AB
3B 88 06 D9
```



Flags 0x23 means client is using SNTPv4 to request data from server

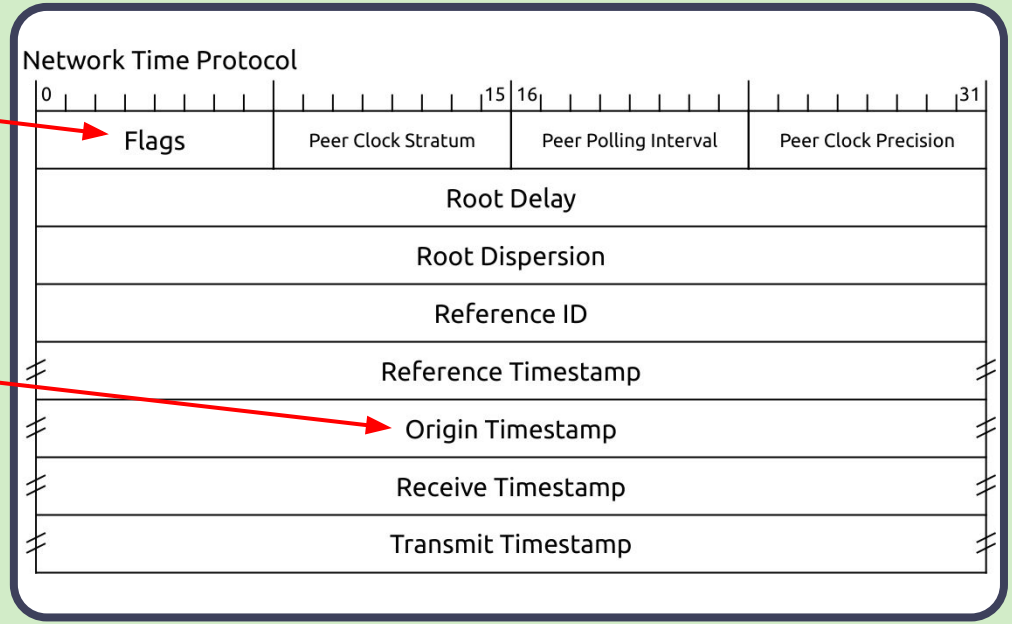
Server to Client Data Frame

```
24 02 00 F7
00 00 00 4C
00 00 00 1B
C9 44 58 6A
E8 83 77 CC
B2 E7 F7 93
E8 83 78 AB
3B 88 06 D9
E8 83 78 AC
23 7C 89 F6
E8 83 78 AC
23 7D A3 A3
```



Server to Client Data Frame

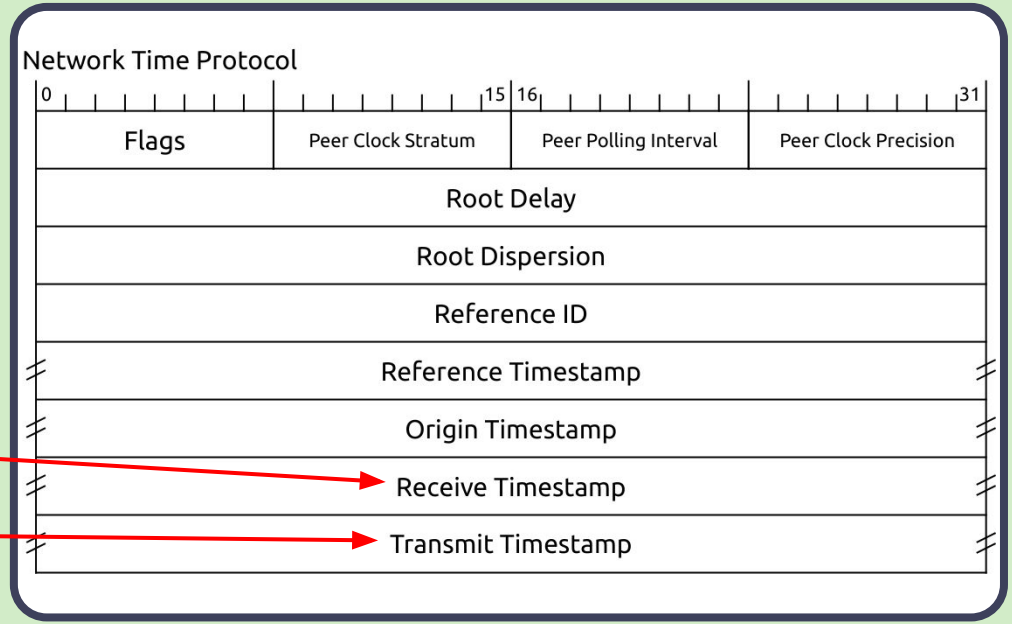
```
24 02 00 F7
00 00 00 4C
00 00 00 1B
C9 44 58 6A
E8 83 77 CC
B2 E7 F7 93
E8 83 78 AB
3B 88 06 D9
E8 83 78 AC
23 7C 89 F6
E8 83 78 AC
23 7D A3 A3
```



Flags 0x24 means server is using SNTPv4 to response data to client

Server to Client Data Frame

```
24 02 00 F7
00 00 00 4C
00 00 00 1B
C9 44 58 6A
E8 83 77 CC
B2 E7 F7 93
E8 83 78 AB
3B 88 06 D9
E8 83 78 AC
23 7C 89 F6
E8 83 78 AC
23 7D A3 A3
```



Flags 0x24 means server is using SNTPv4 to response data to client

Finding Offset and Delay

Offset

$$\theta = \frac{(t_1 - t_0) + (t_2 - t_3)}{2},$$

Round-trip Delay

$$\delta = (t_3 - t_0) - (t_2 - t_1)$$

t0 = Origin Timestamp = 3900930219.232544353
t1 = Receive Timestamp = 3900930220.138619062
t2 = Transmit Timestamp = 3900930220.138635852
t3 = Arrive Timestamp = 3900930220.953471568

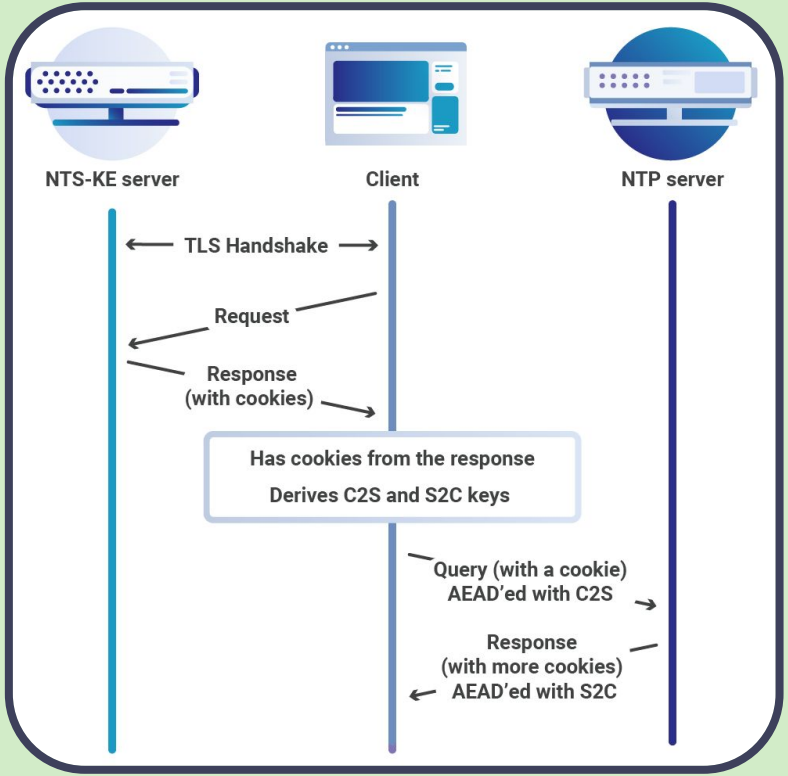
θ = 0.04561924934387207
δ = 1.7209105491638184

According to the results, the delay is 1.72 seconds and the offset is 46 milliseconds. Which means, the client is 46 milliseconds slower than the server.

Arrive Timestamp is the time client receives the response from the server

Security Concerns

- Use of malicious NTP servers can cause client systems to be vulnerable for attacks.
- Misconfigured NTP servers can be used for amplification attacks
- NTP has no authentication therefore can be susceptible to Man-in-the-middle attacks (MITM)
- To Prevent MITM attacks, use Network Time Security (NTS)



END OF PRESENTATION

Reference and credits:

https://en.wikipedia.org/wiki/Network_Time_Protocol

https://www.keycdn.com/support/network-latency

https://blog.cloudflare.com/announcing-cfnts/

INTRODUCING MYSELF

Here is where your presentation begins

TABLE OF CONTENTS

01 MERCURY

Here you could describe the topic of the section

02 VENUS

Here you could describe the topic of the section

03 MARS

Here you could describe the topic of the section

04 JUPITER

Here you could describe the topic of the section

WHOA!

This could be the part of the presentation where you can introduce yourself, write your email...

“This is a quote, words full of wisdom that someone important said and can make the reader get inspired.”

—SOMEONE FAMOUS

01

MY COMPANY

You can enter a subtitle here if you need it

THE SLIDE TITLE GOES RIGHT HERE!

Do you know what helps you make your point clear? Lists like this one:

- They're simple
- You can organize your ideas clearly
- You'll never forget to buy milk!

And the most important thing: the audience won't miss the point of your presentation

WHAT SETS US APART?

MERCURY

Mercury is the closest planet to the Sun and the smallest one in the Solar System

VENUS

Venus has a beautiful name and is the second planet from the Sun. It's terribly hot

OUR OBJECTIVES

MARS

Despite being red, Mars is actually a cold place. It's full of iron oxide dust

JUPITER

It's a gas giant and the biggest planet in the Solar System

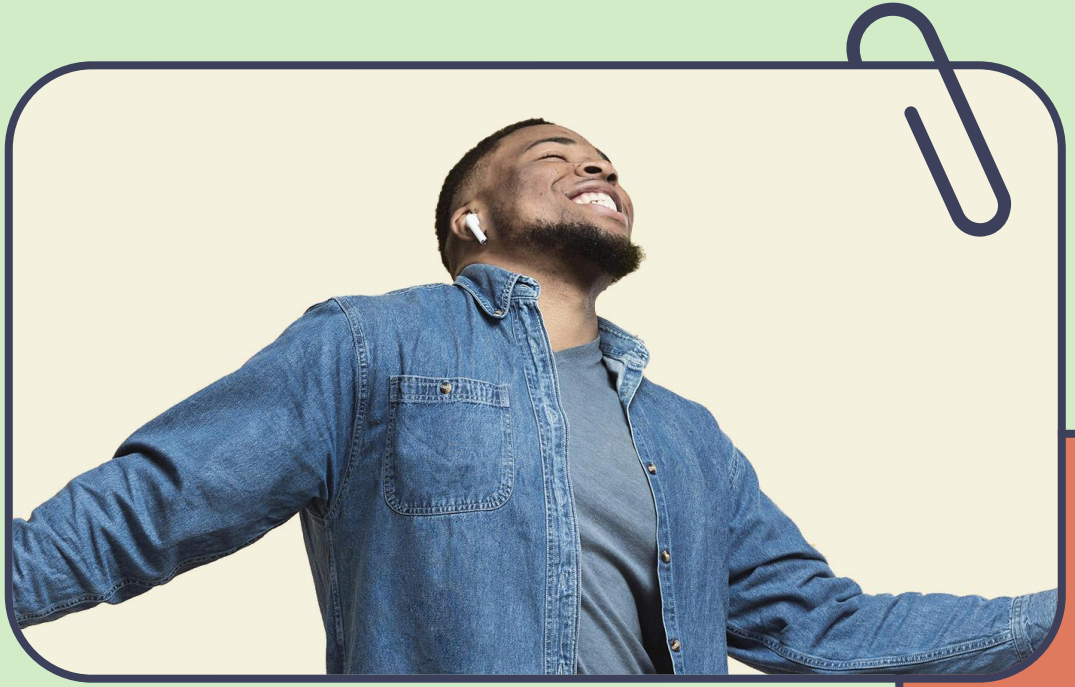
SATURN

Saturn is a gas giant, composed mostly of hydrogen and helium

ABOUT ME

02

Enter a subtitle here if you need it



A PICTURE ALWAYS REINFORCES THE CONCEPT

A PICTURE IS WORTH A
THOUSAND WORDS

AWESOME WORDS

MY TEAM

JOHN

You can replace the image on the screen with your own

JENNA

You can replace the image on the screen with your own

JAMES

You can replace the image on the screen with your own

MY STRENGTHS

MARS

- Despite being red, it's a cold place
- It's full of iron oxide dust
- It's the fourth planet from the Sun
- It was named after a Roman god

PROMOTION

MULTIMEDIA

It’s the biggest planet in the Solar System

PRESS

Saturn is the ringed one and a gas giant

MOBILE APP

Despite being red, Mars is a cold place

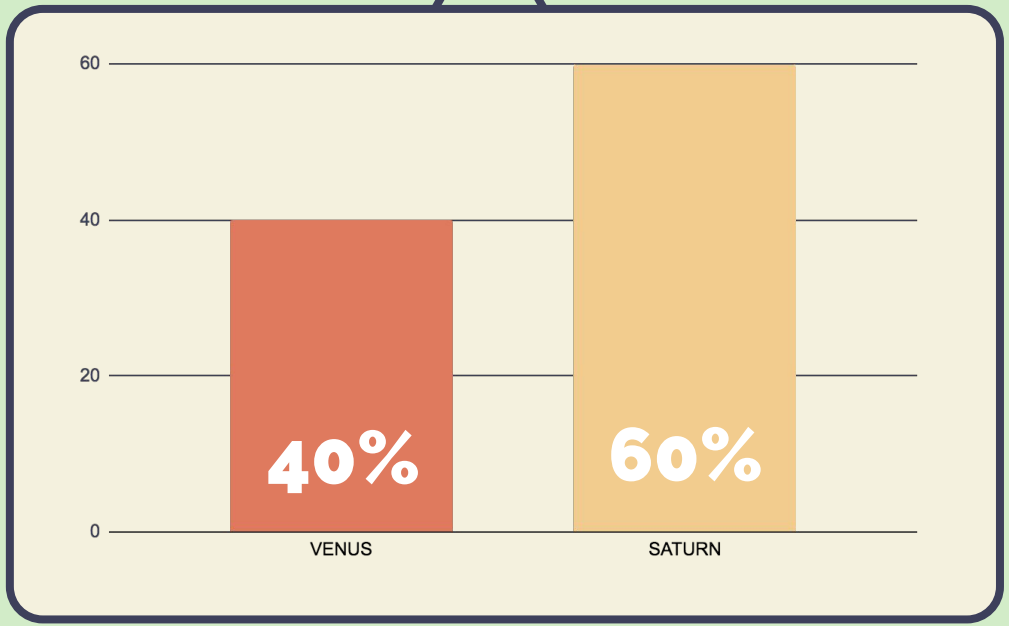
WORK PERFORMANCE

MARS

Despite being red, Mars is actually a cold place

SATURN

Saturn is composed mostly of hydrogen and helium



To modify this graph, click on it, follow the link, change the data and paste the new graph here

WHAT I'VE ACHIEVED

MERCURY

Mercury is the closest planet to the Sun

VENUS

Venus is the second planet from the Sun

MARS

Despite being red, Mars is a cold place

JUPITER

It's the biggest planet in the Solar System

SATURN

Saturn is the ringed one and a gas giant

NEPTUNE

It's the farthest planet from the Sun

MY VALUES

LOYALTY

Neptune is the farthest planet from the Sun

EFFICIENCY

Venus is the second planet from the Sun

COMMITMENT

Jupiter is the biggest planet of them all

PERSONALITY

SATURN

Saturn is the ringed one and a gas giant

VENUS

Venus is the second planet from the Sun

MARS

Despite being red, Mars is actually a cold place

MERCURY

Mercury is the closest planet to the Sun

GOALS

SATURN

Saturn is the ringed one and a gas giant

NEPTUNE

It's the farthest planet from the Sun

JUPITER

It's the biggest planet in the Solar System

SPOKEN LANGUAGES

70%

MERCURY

Mercury is the closest planet to the Sun

30%

JUPITER

It's the biggest planet in the Solar System

To modify this graph, click on it, follow the link, change the data and paste the new graph here

MY EXPERIENCE

COMPANY 1

Despite being red, Mars is actually a cold place

COMPANY 2

It was named after the Roman messenger god

COMPANY 3

It's composed mostly of hydrogen and helium

COMPANY 4

Neptune is the fourth-largest planet of them all

ABOUT MY SKILLS

90%

LEADERSHIP

Despite being red, Mars is actually a cold place

70%

COMMUNICATION

Saturn is composed of hydrogen and helium

MY TABLE

	MASS	DIAMETER	SURFACE
MERCURY	0.06	0.38	0.38
MARS	0.11	0.53	0.38
SATURN	0.06	0.11	95.2

REFERENCES

VENUS

It's the second planet from the Sun

MARS

Despite being red, Mars is a cold place

JUPITER

It's the biggest planet in the Solar System

SATURN

Saturn is the ringed one and a gas giant

NEPTUNE

It's the farthest planet from the Sun

WHERE I'VE BEEN

AMERICA

It's the closest planet to the Sun

EUROPE

Saturn is composed mostly of hydrogen and helium

4,498,300,000

Big numbers catch your audience’s attention

24h

rotation period

333,000.00

earths is the Sun’s mass

386 km

Distance between
Earth and the Moon

INTO THE MEDIA

Mercury is the closest planet to the Sun and the smallest one

DESKTOP SOFTWARE

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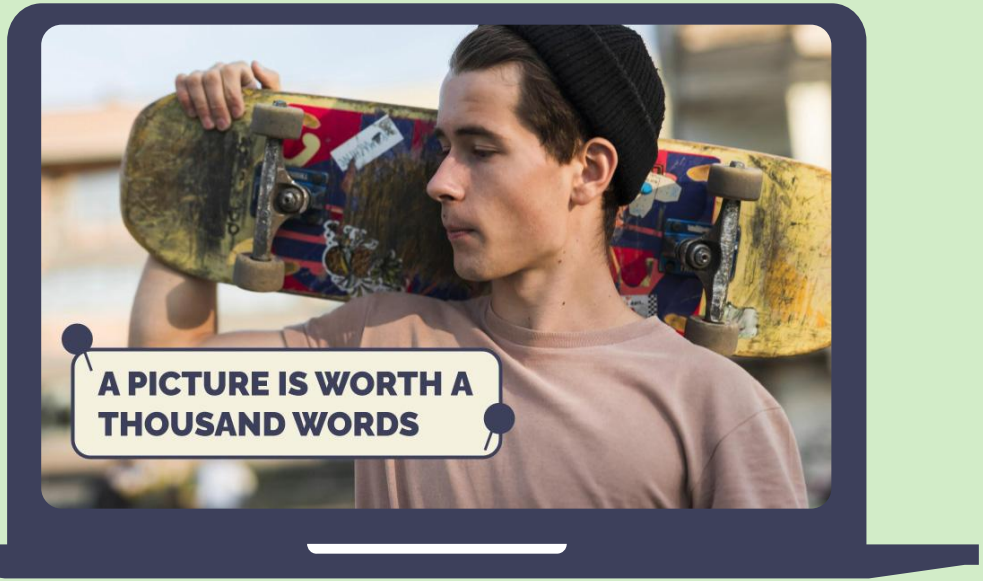

TABLE
T APP

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MOBILE WEB

MY NEW TARGETS

01

MERCURY

Mercury is the closest planet to the Sun and the smallest one in the Solar System

VENUS

Venus has a beautiful name and is the second planet from the Sun. It’s terribly hot

02

Do you have any questions?

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- Close-up female with mobile
- Two female friends
- Medium shot woman
- Hang showing four fingers
- Happy woman listening to music
- Joyful pretty young woman
- Boy expressing his happiness
- Close-up smiley man

VECTORS

- Flat device mockup infographic

RESOURCES

PHOTOS

- Low angle man dancing
- Young man holding a skateboard
- Smiling man
- Happy woman
- Boy expressing his happiness
- Smiley woman
- Portrait woman with sunglasses
- Front view female working

VECTORS

- Device mockup infographic

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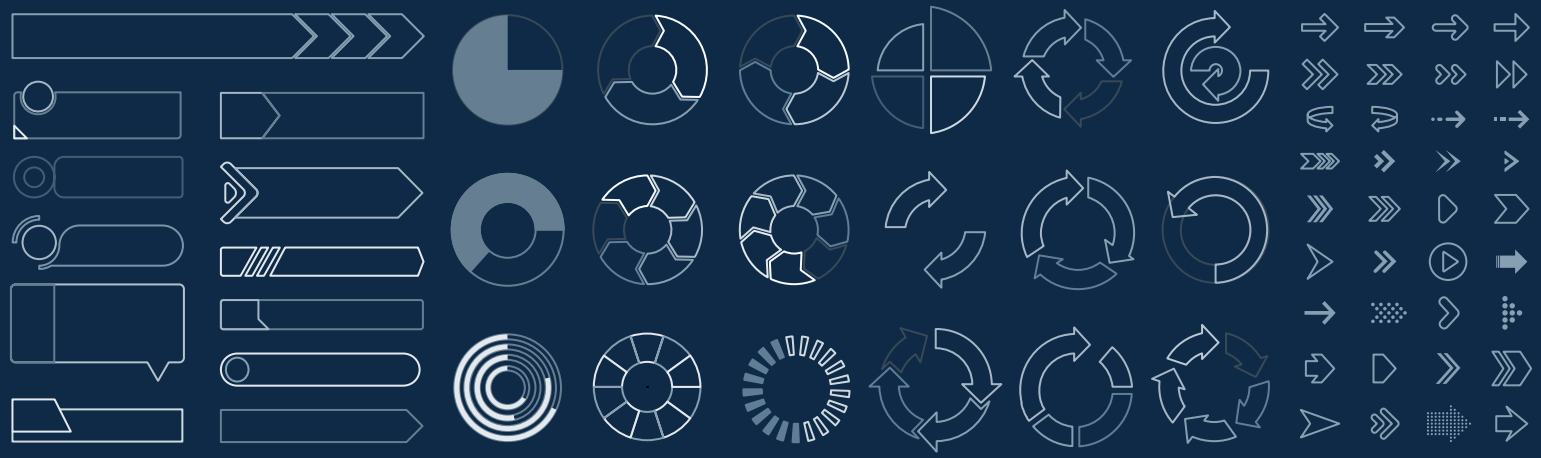

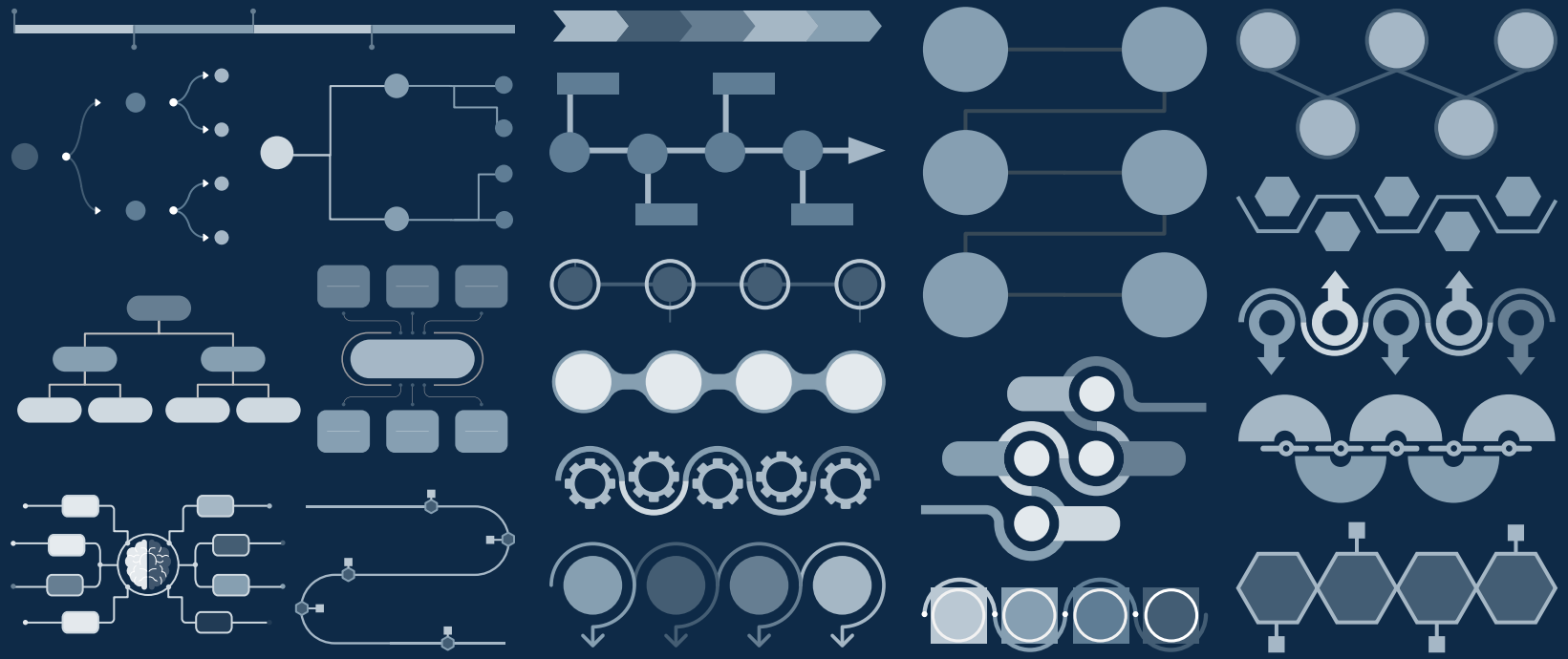

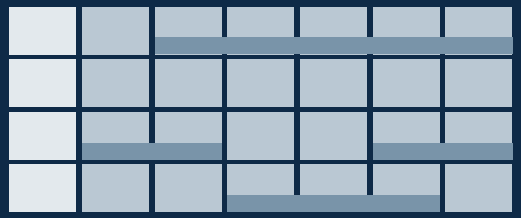

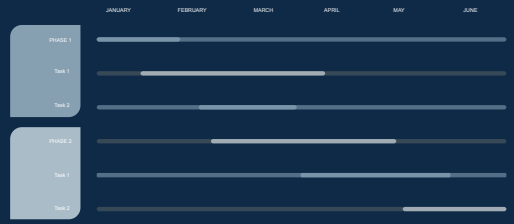
JANUARY
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MAY
JUNE
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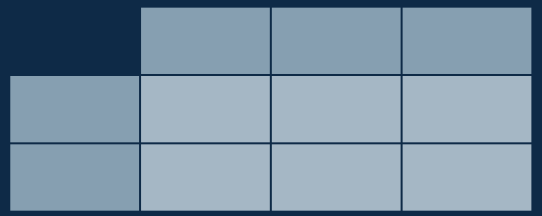
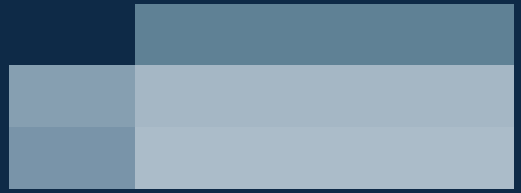
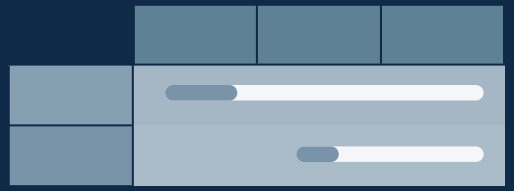
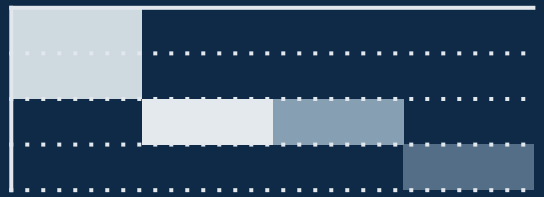
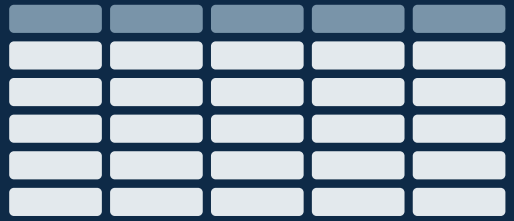
Task 2

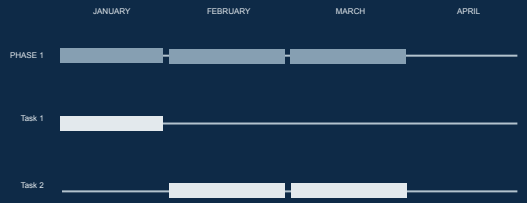











JANUARY
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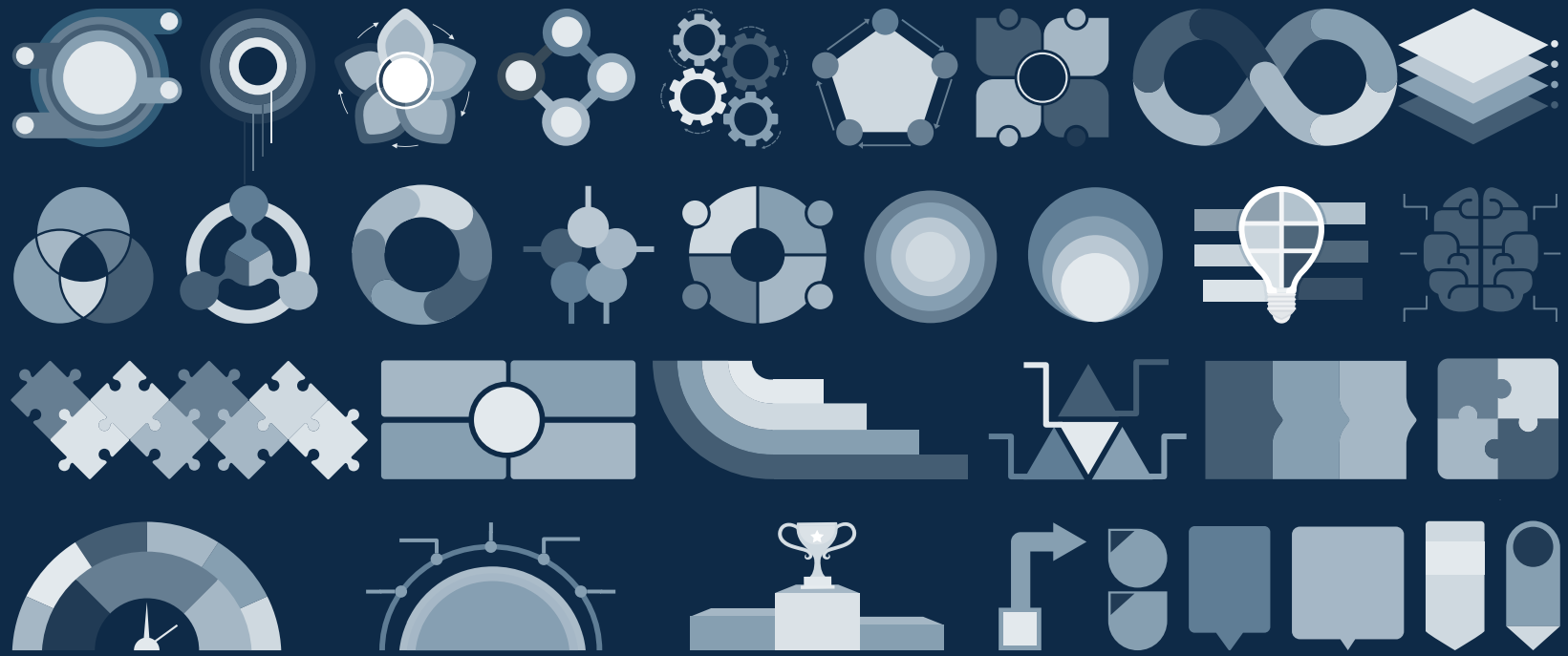

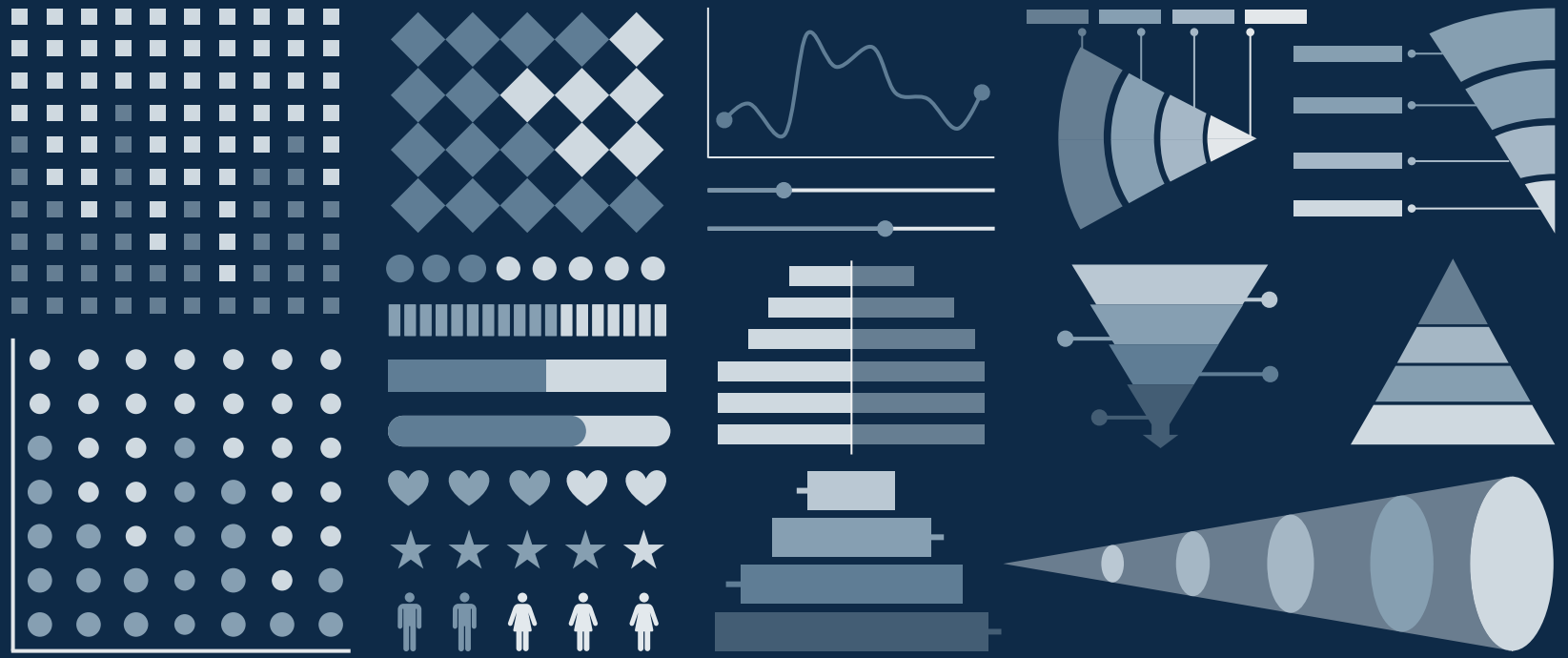

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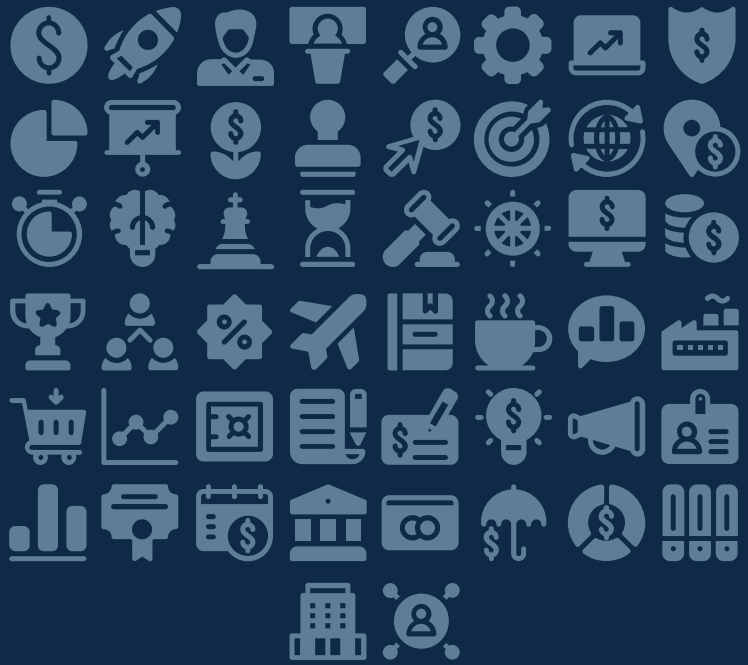
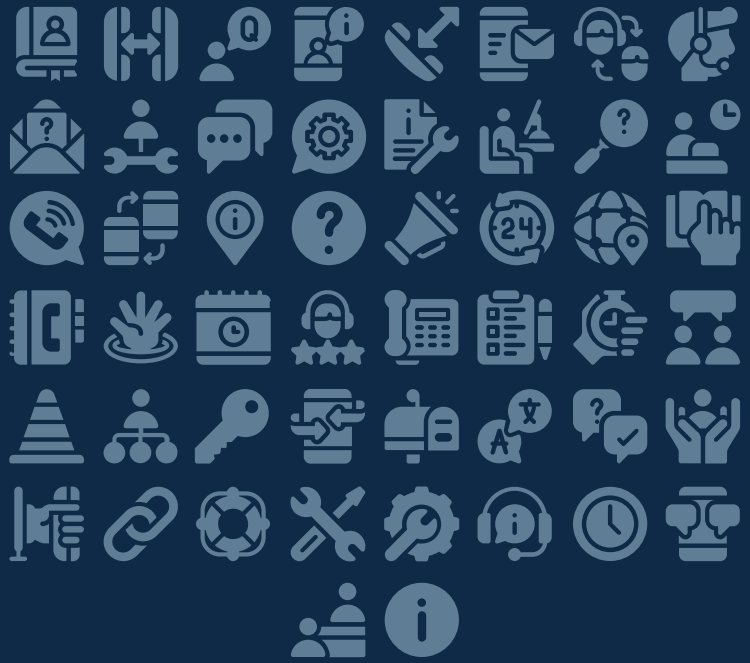
Business Icons

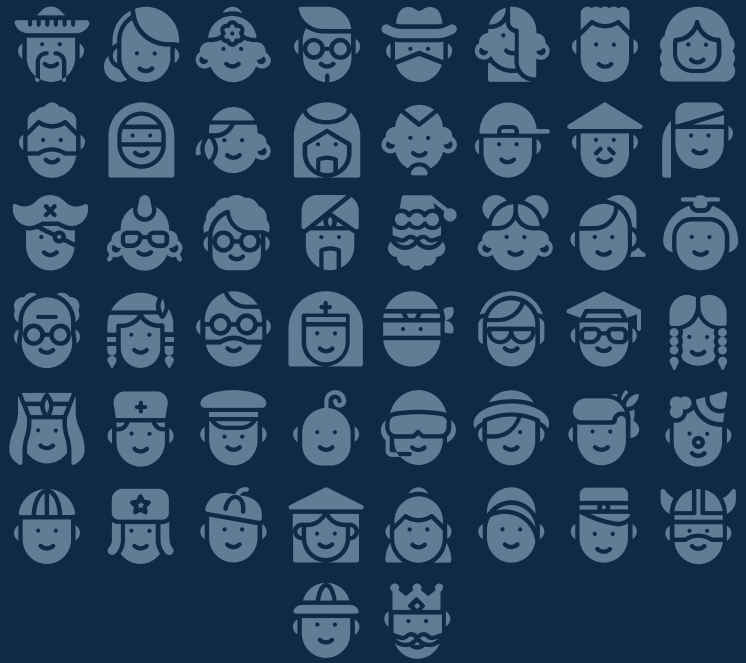


Teamwork Icons

Help & Support Icons



Avatar Icons



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Performing Arts Icons

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